INORGANIC CUME
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A solid knowledge of the synthesis of frequently encountered molecules is extremely valuable to chemists in all areas. For example, you may wish to introduce a small molecule into a reaction under vacuum. In some cases, generation of the molecule inside the apparatus is the best way to proceed. In another case, you may wish to create an isotopically labeled small molecule. With these two examples in mind, provide realistic synthetic schemes for the following. The emphasis here is realistic. This generally, but not always, excludes simply combining elements. Gas cylinders are prohibited.

1. HCl
2. H_2
3. H_2O
4. Cl_2
5. O_2
6. CO_2
7. CO
8. NH_3
9. B_2H_6
10. NO
11. NO_2
12. SO_2
13. N_2
14. C_{60}
15. H_2C_2